

Curriculum Vitae of Nian-Feng Tzeng

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PROFESSIONAL EXPERIENCE

Professor of Computer Engineering and Director of Computer Architecture and Networks Laboratory (January 1999 - Present), Center for Advanced Computer Studies, University of Louisiana at Lafayette.

Visiting Professor (August - December 2000), Sabbatical with Department of Computer Science, Rice University.

Professor of Computer Engineering (August 1997 - December 1998), Center for Advanced Computer Studies, University of Louisiana at Lafayette.

Senior Faculty Fellow (June - August 1997), Naval Surface Warfare Center, U.S. Navy-ASEE Summer Faculty Research Program.

Associate Professor of Computer Engineering (1992 - 1997), Center for Advanced Computer Studies, University of Louisiana at Lafayette.

Assistant Professor of Computer Engineering (1987 - 1992), Center for Advanced Computer Studies, University of Louisiana at Lafayette.

Member of Technical Staff (1986 - 1987), AT&T Bell Labs, Columbus, Ohio.

EDUCATION

Ph.D., University of Illinois at Urbana-Champaign, 1986, Computer Science.

M.S., National Taiwan University, 1980, Electrical and Computer Engineering.

B.S., National Chiao Tung University, Taiwan, 1978, Computer Science.

RESEARCH AREAS

Computer Communications and Networks
Software System Support for Cluster Computing
High-Performance Computer Systems
Parallel and Distributed Processing
Fault-Tolerant Computing

RESEARCH GRANTS AND CONTRACTS — Summary

- Total Amount: \$3,644,409 (university matching excluded)

- Sponsors: National Science Foundation, State of Louisiana, Louisiana Board of Regents, Army Research Office.

List of Research Grants and Contracts

17. National Science Foundation, Division of Computer-Communications Research, "Architectural Support for Scalable High-Speed Routers," \$300,000, September 2001 through August 2005.
16. State of Louisiana, Information Technology Initiative, "Scalable Routers," \$1,727,000, September 2001 through June 2006 (jointly with M. A. Bayoumi).
15. National Science Foundation, Division of Computer-Communications Research, "Reconfiguration and Performance Issues in Software Distributed Shared Memory Systems," \$212,065, September 1998 through August 2002.
14. National Science Foundation, Major Research Instrumentation Program, "Acquisition of Networked Heterogeneous Computer Systems," \$102,400, September 1998 through August 2002.
13. BORSEF, Board of Regents, State of Louisiana, "Instrumentation for Research on Communications and Networks," \$50,000, June 2000 through June 2001.
12. Army Research Office, Electronics Division, "High-Performance Wireless Information Systems," \$229,504, May 1998 through April 2001 (jointly with G.-L. Feng).
11. BORSEF, Board of Regents, State of Louisiana, "High-Performance Wireless Information Systems," \$97,292, October 1998 through April 2001.
10. BORSEF, Board of Regents, State of Louisiana, "Acquisition of Networked Computer Systems," \$60,300, June 1998 through June 1999.
9. LEQSF, Board of Regents, State of Louisiana, "Interoperable Underground Wavelet Transients Munitions Classification," \$150,000, June 1996 through June 1999 (jointly with H. Szu and M. A. Bayoumi).
8. National Science Foundation, Computer and Computation Research, "Investigation into Faulty and Incomplete Message-Passing Parallel Computers," \$60,030, August 1993 through July 1996.
7. LEQSF, Board of Regents, State of Louisiana, "The Design and Development of Next Generation Geographic Information Systems," \$63,000, June 1994 through June 1996 (jointly with N. Pissinou).
6. LaSER, NSF and Board of Regents, State of Louisiana, "Developing an Architectural Framework for Distributed GIS," \$55,000, June 1994 through June 1995 (jointly with N. Pissinou and R. Logan).
5. LEQSF, Board of Regents, State of Louisiana, "Parallelization for Finite-Element Computation," \$65,050, June 1993 through June 1995 (jointly with H.-C. Chen).
4. National Science Foundation, Microelectronic Information Processing Systems, "Improving the Communication Performance of Multiprocessors," \$98,015, May 1992 through April 1995.
3. LEQSF, Board of Regents, State of Louisiana, "Communication Performance Improvement in Multiprocessors under Nonuniform Traffic," \$55,597, June 1992 through June 1994.
2. LEQSF, Board of Regents, State of Louisiana, "Prototyping of Parallel Algorithms for Engineering Applications," \$150,000, June 1992 through June 1993 (jointly with M. A. Bayoumi and M. Montenyohl).
1. National Science Foundation, Microelectronic Information Processing Systems, "Yield and Reliability Considerations of VLSI/WSI Computing Systems," \$169,206, September 1988 through August 1990 (jointly with L.N. Bhuyan and T.R. Rao).

PUBLICATIONS

Refereed Journal Papers

The following publications appeared in leading archival journals after fully peer-reviewed.

27. S. S. Fu, N.-F. Tzeng and J.-Y. Chung, "Empirical Evaluation of Mutual Exclusion Algorithms for Distributed Systems," *Journal of Parallel and Distributed Computing*, vol. 60, pp. 785-806, July 2000.
26. N.-F. Tzeng and H.-L. Chen, "Fast Compaction in Hypercubes," *IEEE Transactions on Parallel and Distributed Systems*, vol. 9, pp. 50-56, Jan. 1998.
25. H.-L. Chen and N.-F. Tzeng, "A Boolean Expression-Based Approach for Maximum Incomplete Subcube Identification in Faulty Hypercubes," *IEEE Transactions on Parallel and Distributed Systems*, vol. 8, pp. 1171-1183, Nov. 1997.
24. H.-L. Chen and N.-F. Tzeng, "Subcube Determination in Faulty Hypercubes," *IEEE Transactions on Computers*, vol. 46, pp. 871-879, Aug. 1997.
23. S. S. Fu and N.-F. Tzeng, "A Circular List-Based Mutual Exclusion Scheme for Large Shared-Memory Multiprocessors," *IEEE Transactions on Parallel and Distributed Systems*, vol. 8, pp. 628-639, June 1997.
22. H.-L. Chen and N.-F. Tzeng, "On-Line Task Migration in Hypercubes through Double Disjoint Paths," *IEEE Transactions on Computers*, vol. 46, pp. 379-384, Mar. 1997.
21. N.-F. Tzeng and G. Lin, "Efficient Determination of Maximum Incomplete Subcubes in Hypercubes with Faults," *IEEE Transactions on Computers*, vol. 45, pp. 1303-1308, Nov. 1996.
20. N.-F. Tzeng and H. Kumar, "Traffic Analysis and Simulation Performance of Incomplete Hypercubes," *IEEE Transactions on Parallel and Distributed Systems*, vol. 7, pp. 740-754, July 1996.
19. N.-F. Tzeng and G.-L. Feng, "Resource Allocation in Cube Network Systems Based on the Covering Radius," *IEEE Transactions on Parallel and Distributed Systems*, vol. 7, pp. 328-342, Apr. 1996.
18. G. Lin and N.-F. Tzeng, "Effective Utilization of Hypercubes in the Presence of Faults," *Journal of Parallel and Distributed Computing*, vol. 32, pp. 223-231, Feb. 1996.
17. N.-F. Tzeng and H.-L. Chen, "Structural and Tree Embedding Aspects of Incomplete Hypercubes," *IEEE Transactions on Computers*, vol. 43, pp. 1434-1439, Dec. 1994.
16. N.-F. Tzeng, "Reliable Butterfly Distributed-Memory Multiprocessors," *IEEE Transactions on Computers*, vol. 43, pp. 1004-1013, Sept. 1994.
15. N.-F. Tzeng and P.-J. Chuang, "A Pairwise Substitutional Fault Tolerance Technique for the Cube-Connected Cycles Architecture," *IEEE Transactions on Parallel and Distributed Systems*, vol. 5, pp. 433-439, Apr. 1994.
14. P.-J. Chuang and N.-F. Tzeng, "Allocating Precise Submeshes in Mesh Connected Systems," *IEEE Transactions on Parallel and Distributed Systems*, vol. 5, pp. 211-217, Feb. 1994.
13. H.-L. Chen and N.-F. Tzeng, "Efficient Resource Placement in Hypercubes Using Multiple-Adjacency Codes," *IEEE Transactions on Computers*, vol. 43, pp. 23-33, Jan. 1994.
12. N.-F. Tzeng, P.-J. Chuang, and C.-H. Wu, "Creating Disjoint Paths in Gamma Interconnection Networks," *IEEE Transactions on Computers*, vol. 42, pp. 1247-1252, Oct. 1993.
11. N.-F. Tzeng, "A Reliable Cube-Connected Cycles Structure," *Journal of Parallel and Distributed Computing*, vol. 19, pp. 64-71, Sept. 1993.

10. N.-F. Tzeng, "Reconfiguration and Analysis of a Fault-Tolerant Circular Butterfly Parallel System," *IEEE Transactions on Parallel and Distributed Systems*, vol. 4, pp. 855-863, Aug. 1993.
9. N.-F. Tzeng, "A Cube-Connected Cycles Architecture with High Reliability and Improved Performance," *IEEE Transactions on Computers*, vol. 42, pp. 246-253, Feb. 1993.
8. N.-F. Tzeng, "A Cost-Effective Combining Structure for Large-Scale Shared-Memory Multiprocessors," *IEEE Transactions on Computers*, vol. 41, pp. 1420-1429, Nov. 1992.
7. P.-J. Chuang and N.-F. Tzeng, "A Fast Recognition-Complete Processor Allocation Strategy for Hypercube Computers," *IEEE Transactions on Computers*, vol. 41, pp. 467-479, Apr. 1992.
6. N.-F. Tzeng and H.-L. Chen, "An Effective Approach to the Enhancement of Incomplete Hypercubes," *Journal of Parallel and Distributed Computing*, vol. 14, pp. 163-174, Feb. 1992.
5. S. K. Mishra, V. V. Raghavan, and N.-F. Tzeng, "Efficient Algorithms for Selection of Recovery Points in Tree Task Models," *IEEE Transactions on Software Engineering*, vol. 17, pp. 731-734, July 1991.
4. N.-F. Tzeng, "Alleviating the Impact of Tree Saturation on Multistage Interconnection Network Performance," *Journal of Parallel and Distributed Computing*, special issue on shared-memory multiprocessors, pp. 107-117, June 1991.
3. N.-F. Tzeng and S. Wei, "Enhanced Hypercubes," *IEEE Transactions on Computers*, vol. 40, pp. 284-294, Mar. 1991.
2. N.-F. Tzeng, P.-C. Yew, and C.-Q. Zhu, "Realizing Fault-Tolerant Interconnection Networks Via Chaining," *IEEE Transactions on Computers*, special issue on fault-tolerant computing, vol. 37, pp. 458-462, Apr. 1988.
1. P.-C. Yew, N.-F. Tzeng, and D. H. Lawrie, "Distributing Hot-Spot Addressing in Large-Scale Multiprocessors," *IEEE Transactions on Computers*, special issue on parallel and distributed computing, vol. C-36, pp. 388-395, Apr. 1987.

Work in Progress

- N.-F. Tzeng and R. Batchu, "Scalable Switching Fabrics for High-Performance Routers," submitted to *IEEE/ACM Transactions on Networking* for publication.

Book Articles

- N.-F. Tzeng, P.-C. Yew, and C.-Q. Zhu, "Realizing Fault-Tolerant Interconnection Networks Via Chaining," *Advances in Distributed System Reliability*, Ed. by S. Rai and D. P. Agrawal, IEEE Computer Society Press, 1990, pp. 67-71.

Refereed Papers in Top Conference Proceedings

The following publications appeared in top quality conference proceedings after rigorously peer-reviewed. They normally had acceptance rates of no more than 33%.

34. K. Vibhatavani, N.-F. Tzeng, and A. Kongmunvattana, "Simultaneous Multithreading-Based Routers," *Proc. 29th Int'l Conf. on Parallel Processing*, Aug. 2000, pp. 362-369.
33. A. Kongmunvattana, S. Tanchatchawal, and N.-F. Tzeng, "Coherence-Based Coordinated Checkpointing for Software Distributed Shared Memory Systems," *Proc. 20th Int'l Conf. on Distributed Computing Systems*, Apr. 2000, pp. 556-563.
32. A. Kongmunvattana and N.-F. Tzeng, "Harmonious Logging for Adaptive Software Distributed Shared Memory Systems," *Proc. 18th IEEE Symp. Reliable Distributed Systems (SRDS)*, Oct. 1999, pp. 202-211.

31. A. Kongmunvattana and N.-F. Tzeng, "Coherence-Centric Logging and Recovery for Home-Based Software Distributed Shared Memory," *Proc. 1999 Int'l Conf. on Parallel Processing*, Sept. 1999, pp. 274-281.
30. A. Kongmunvattana and N.-F. Tzeng, "Lazy Logging and Prefetch-Based Crash Recovery in Software Distributed Shared Memory Systems," *Proc. 1999 IEEE Int'l Parallel Processing Symposium*, April 1999, pp. 399-406.
29. N.-F. Tzeng and A. Kongmunvattana, "Distributed Shared Memory Systems with Improved Barrier Synchronization and Data Transfer," *Proc. 11th ACM Int'l Conf. Supercomputing*, July 1997, pp. 148-155.
28. S. S. Fu and N.-F. Tzeng, "Aggressive Release Consistency for Software Distributed Shared Memory," *Proc. 17th Int'l Conf. Distributed Computing Systems*, May 1997, pp. 288-295.
27. S. S. Fu, N.-F. Tzeng, and Z. Li, "Empirical Evaluation of Distributed Mutual Exclusion Algorithms," *Proc. 11th Int'l Parallel Processing Symposium*, Apr. 1997, pp. 255-259.
26. N.-F. Tzeng and G. Lin, "Maximum Reconfiguration of 2-D Mesh Systems with Faults," *Proc. 1996 Int'l Conf. Parallel Processing*, Aug. 1996, vol. I, pp. 77-84.
25. N.-F. Tzeng and S. Fu, "Efficient Implementation of Mutual Exclusion Locks in Large Multiprocessors," *Proc. 9th Int'l Parallel Processing Symposium*, Apr. 1995, pp. 270-275.
24. G. Lin and N.-F. Tzeng, "Reconfiguration and Experiments in a Faulty Hypercube," *Proc. 9th Int'l Parallel Processing Symposium*, Apr. 1995, pp. 592-598.
23. H.-L. Chen and N.-F. Tzeng, "Speedy Task Migration in Hypercube Systems," *Proc. 23rd Int'l Conf. Parallel Processing*, Aug. 1994, vol. III, pp. 74-78.
22. H.-L. Chen and N.-F. Tzeng, "Distributed Identification of All Maximal Incomplete Subcubes in a Faulty Hypercube," *Proc. 8th Int'l Parallel Processing Symposium*, Apr. 1994, pp. 723-728.
21. N.-F. Tzeng, "Empirical Evaluation of Incomplete Hypercube Systems," *Proc. 22nd Int'l Conf. Parallel Processing*, Aug. 1993, vol. I, pp. 96-99.
20. N.-F. Tzeng and G.-L. Feng, "On Resource Allocation in Binary n -Cube Network Systems," *Proc. 22nd Int'l Conf. Parallel Processing*, Aug. 1993, vol. II, pp. 209-212.
19. H.-L. Chen and N.-F. Tzeng, "Quick Determination of Subcubes in a Faulty Hypercube," *Proc. 21st Int'l Conf. Parallel Processing*, Aug. 1992, vol. III, pp. 338-345.
18. N.-F. Tzeng, "Design of a Highly Reliable Cube-Connected Cycles Architecture," *Proc. ACM/IEEE Supercomputing '91*, Nov. 1991, pp. 776-785.
17. N.-F. Tzeng, "An Approach to the Performance Improvement of Multistage Interconnection Networks with Nonuniform Traffic Spots," *Proc. 20th Int'l Conf. Parallel Processing*, Aug. 1991, vol. I, pp. 542-545.
16. H.-L. Chen and N.-F. Tzeng, "Fault-Tolerant Resource Placement in Hypercube Computers," *Proc. 20th Int'l Conf. Parallel Processing*, Aug. 1991, vol. I, pp. 517-524.
15. N.-F. Tzeng, "A Reliable Butterfly Network for Distributed-Memory Multiprocessors," *Proc. 20th Int'l Conf. Parallel Processing*, Aug. 1991, vol. I, pp. 323-326.
14. N.-F. Tzeng and P.-J. Chuang, "Fault-Tolerant Gamma Interconnection Networks," *Proc. 21st IEEE Int'l Symp. Fault-Tolerant Computing*, June 1991, pp. 282-289.
13. P.-J. Chuang and N.-F. Tzeng, "An Efficient Submesh Allocation Strategy for Mesh Computer Systems," *Proc. 11th IEEE Int'l Conf. Distributed Computing Systems*, May 1991, pp. 256-263.
12. C.-H. Chen and N.-F. Tzeng, "An Area-Efficient Reconfigurable Binary Tree Architecture," *Proc.*

1990 *IEEE Int'l Conf. Computer Design — VLSI in Computers and Processors*, Sept. 1990, pp. 376-379.

11. N.-F. Tzeng, S. Bhattacharya, and P.-J. Chuang, "Fault-Tolerant Cube-Connected Cycles Structures through Dimensional Substitution," *Proc. 1990 Int'l Conf. Parallel Processing*, Aug. 1990, vol. I, pp. 433-440.
10. N.-F. Tzeng, P.-J. Chuang, and H.-L. Chen, "Embeddings in Incomplete Hypercubes," *Proc. 1990 Int'l Conf. Parallel Processing*, Aug. 1990, vol. III, pp. 335-339.
9. N.-F. Tzeng, "Analysis of a Variant Hypercube Topology," *Proc. 1990 ACM Int'l Conf. Supercomputing*, June 1990, pp. 60-70.
8. P.-J. Chuang and N.-F. Tzeng, "Dynamic Processor Allocation in Hypercube Computers," *Proc. 17th IEEE/ACM Int'l Symp. Computer Architecture*, May 1990, pp. 40-49.
7. N.-F. Tzeng, "Structural Properties of Incomplete Hypercube Computers," *Proc. 10th IEEE Int'l Conf. Distributed Computing Systems*, May 1990, pp. 262-269 (the **best paper** out of 331 submissions).
6. N.-F. Tzeng, "Design of a Novel Combining Structure for Shared-Memory Multiprocessors," *Proc. 1989 Int'l Conf. Parallel Processing*, Aug. 1989, pp. I 1-8.
5. H.-L. Chen and N.-F. Tzeng, "Enhanced Incomplete Hypercubes," *Proc. 1989 Int'l Conf. Parallel Processing*, Aug. 1989, vol. I, pp. 270-277.
4. P.-C. Yew, N.-F. Tzeng, and D. H. Lawrie, "Distributing Hot-Spot Addressing in Large-Scale Multiprocessors," *Proc. 1986 IEEE Int'l Conf. Parallel Processing*, Aug. 1986, pp. 51-58.
3. N.-F. Tzeng, P.-C. Yew, and C.-Q. Zhu, "Fault-Diagnosis in a Multiple-Path Interconnection Network," *Proc. 16th IEEE Int'l Symp. Fault-Tolerant Computing*, July 1986, pp. 98-103.
2. N.-F. Tzeng, P.-C. Yew, and C.-Q. Zhu, "The Performance of a Fault-Tolerant Multistage Interconnection Network," *Proc. 1985 IEEE Int'l Conf. Parallel Processing*, Aug. 1985, pp. 458-465.
1. N.-F. Tzeng, P.-C. Yew, and C.-Q. Zhu, "A Fault-Tolerant Scheme for Multistage Interconnection Networks," *Proc. 12th IEEE/ACM Int'l Symp. Computer Architecture*, June 1985, pp. 368-375.

Refereed Papers in Other Conference Proceedings

The following papers were reviewed and presented in conferences, whose accepted rates typically ranged from 33% to 50%.

13. A. Ganjoo and N.-F. Tzeng, "Influence of High-Level Program Structures on Branch Prediction Accuracy," *Proc. Euromicro Conf. on Digital System Design*, Sept. 2000, vol. 1, pp. 316-322.
12. N.-F. Tzeng, K. Ponnuru, and K. Vibhatavani, "A Cost-Effective Design for ATM Switching Fabrics," *Proc. 1999 IEEE Int'l Conference on Communications (ICC '99)*, June 1999, pp. S37.4.1-5.
11. S. Fu and N.-F. Tzeng, "Lock Improvement Scheme for Release Consistency in Distributed Shared Memory Systems," *Proc. 6th Symp. Frontiers of Massively Parallel Computation*, Oct. 1996, pp. 255-262.
10. Y. Liao, M. Lu, N.-F. Tzeng, "The Parilndrome Network for Fault-Tolerant Interconnection," *Proc. 8th IEEE Symp. Parallel and Distributed Processing*, Oct. 1996, pp. 556-561.
9. C.-H. Wu, F. F. Young, and N.-F. Tzeng, "Shortening Turnaround Time for Two-Dimensional Plasma Processing Simulations via Vectorization and Parallelization Techniques," *Proc. 7th SIAM Conf. Parallel Processing for Scientific Computing*, Feb. 1995, pp. 237-238.
8. H.-L. Chen and N.-F. Tzeng, "Task Migration in Hypercubes Using All Disjoint Paths," *Proc. 7th*

Int'l Conf. Parallel and Distributed Computing Systems, Oct. 1994, pp. 358-363.

7. N.-F. Tzeng and G. Lin, "Identifying Maximal Incomplete Subcubes in Faulty Hypercubes," *Proc. 7th Int'l Conf. Parallel and Distributed Computing Systems*, Oct. 1994, pp. 186-193.
6. N.-F. Tzeng and H. Kumar, "Analysis of Link Traffic in Incomplete Hypercubes," *Proc. 5th IEEE Symp. Parallel and Distributed Processing*, Dec. 1993, pp. 312-319.
5. N.-F. Tzeng, "An Alternative Combining Architecture for Large-Scale Shared-Memory Multiprocessors," *Proc. 1991 IEEE Int'l Symp. Circuits and Systems*, June 1991, pp. 1049-1052.
4. N.-F. Tzeng, "A Reconfigurable Cube-Connected Cycles Architecture for Wafer Scale Integration," *Proc. IEEE Int'l Conf. Wafer Scale Integration*, Jan. 1991, pp. 33-39.
3. N.-F. Tzeng and C.-H. Chen, "Yield Evaluation of WSI Parallel Systems," *Proc. IEEE Int'l Conf. Wafer Scale Integration*, Jan. 1991, pp. 156-162.
2. N.-F. Tzeng, "An Effective Fault-Tolerant Technique for Circular Butterfly Parallel Systems," *Proc. 2nd IEEE Symp. Parallel & Distributed Processing*, Dec. 1990, pp. 223-226.
1. N.-F. Tzeng and S. Bhattacharya, "Binary Tree Layouts with Adequate I/O Support," *Proc. 1990 IEEE Int'l Symp. Circuits and Systems*, May 1990.

Conference Submissions under Review

- N.-F. Tzeng and K. Vibhatavanij, "Cache-Oriented Multistage Structure for Fast Packet Forwarding in Scalable Routers," submitted for possible presentation.
- N.-F. Tzeng and R. Batchu, "Design and Evaluation of Scalable Switching Fabrics for High-Performance Routers," submitted for possible presentation.
- N.-F. Tzeng and M. Mandviwalla, "Cost-Effective Switching Fabrics with Distributed Control for Scalable Routers," submitted for possible presentation.

Other Articles and Edited Volumes

- N.-F. Tzeng, "Message from the Chair," *Newsletter of the Technical Committee on Distributed Processing*, IEEE Computer Society, Spring 2001.
- N.-F. Tzeng, "Message from the Chair," *Newsletter of the Technical Committee on Distributed Processing*, IEEE Computer Society, Fall 2000.
- N.-F. Tzeng, "Message from the Chair," *Newsletter of the Technical Committee on Distributed Processing*, IEEE Computer Society, Spring 2000, pp. 3.
- H. Szu, N.-F. Tzeng, and G.-L. Feng, "Wireless Subnetworks across the Gulf of Mexico for the Next Generation Internet (NGI)," *Workshop on Research Directions for the Next Generation Internet* (sponsored by NSF, ARPA, and other Federal agencies), May 1997.
- N.-F. Tzeng, Ed., *Newsletter of the Technical Committee on Distributed Processing*, IEEE Computer Society, Summer 1997, 24 pages.
- N.-F. Tzeng and S. J. Wallach, "Issues on the Architecture and the Design of Distributed Shared Memory Systems," *IEEE Int'l Conf. Computer Design: VLSI in Computers & Processors (ICCD '96)*, Oct. 1996, pp. 60-61.
- J. Urban and N.-F. Tzeng, Technical Activities Forum on "Distributed Processing," *Computer Magazine*, June 1996, pp. 99.
- N.-F. Tzeng and P.-C. Yew, Ed., "Special Issue on Distributed Shared Memory Systems," *Journal of Parallel and Distributed Computing*, vol. 29, Sept. 1995, 131 pages.

- N.-F. Tzeng, Review of C. Wu and M. Lee's article "Performance Analysis of Multistage Interconnection Network Configurations and Operations," (published in *IEEE Trans. Computers*, pp. 18-27, Jan. 1992), *Computing Reviews*, Aug. 1993.
- N.-F. Tzeng, Review of D. Rau, J. A. Fortes, and H. J. Siegel's article "Destination Tag Routing Techniques Based on a State Model for the IADM Network," (published in *IEEE Trans. Computers*, pp. 274-285, Mar. 1992), *Computing Reviews*, Apr. 1993.
- N.-F. Tzeng, Review of K. Padmanabhan's article "Cube Structures for Multiprocessors," (published in *Commun. ACM*, pp. 43-52, Jan. 1990), *Computing Reviews*, Mar. 1991.

PROFESSIONAL HONORS AND ACTIVITIES

- Recipient, University Foundation Distinguished Professor Award, 1997.
- Recipient, the Outstanding Paper Award of 10th IEEE International Conference on Distributed Computing Systems, May 1990.
- Chair, Technical Committee on Distributed Processing, IEEE Computer Society (Elected, 1999 - Present).
- Editor, IEEE Transactions on Parallel and Distributed Systems (1998 - Present).
- Editor, IEEE Transactions on Computers (1994 - 1998).
- Co-Guest Editor, special issue of the Journal of Parallel and Distributed Computing on distributed shared memory systems, September 1995.
- Panelist for Proposal Evaluation, NSF Operating Systems and Compilers Program, Division of Computer-Communications Research, 2000 and January 2002.
- Panelist for Proposal Evaluation, NSF Information Technology Research (ITR) Program, 2000.
- Panelist for Proposal Evaluation, NSF Computer Systems Architecture Program, Division of Computer-Communications Research, 1999 and February 2002.
- Panelist for Proposal Evaluation, NSF POWRE Program, Division of Experimental and Integrative Activities, 1999.
- Distinguished Visitor, IEEE Computer Society (1994 - 1997).
- Newsletter Editor, Technical Committee on Distributed Processing, IEEE Computer Society (1995 - 1997).
- Senior Member, IEEE (Computer Society).
- Member, ACM (SIGARCH, SIGCOMM).
- Member, Tau Beta Pi Honorary Society.
- Listed in Who's Who in the World, Marquis Who's Who, 12th Edition, 1995.
- Workshops Chair, 2002 Symposium on Applications and the Internet (SAINT-2002), January - February 2002.
- Tutorials Chair, 2002 IEEE International Conference on Parallel and Distributed Systems (ICPAD-2002), December 2002.
- Member, technical program committee of 20th IEEE Symposium on Reliable Distributed Systems (SRDS), October 2001.
- Member, technical program committee of 2001 International Conference on Computer Networks and Mobile Computing, October 2001.
- Member, technical program committee of 5th International Conference on Computer Science and Informatics (CS&I), February 2000.
- Member, technical program committee of International Conference on High Performance Computing (HiPC'99), December 1999.

- Member, technical program committee of International Conference on Parallel and Distributed Systems (ICPADS'98), December 1998.
- Member, technical program committee of International Conference on Computer Design, October 1997.
- Member, technical program committee of International Conference on Computer Design — VLSI in Computers and Processors, October 1996.
- Member, technical program committee of International Conference on High Performance Computing, December 1995.
- Member, technical program committee of 9th ACM International Conference on Supercomputing, July 1995.
- Member, technical program committee of 1st International Symposium on High Performance Computer Architecture, January 1995.
- Member, technical program committee of 7th International Conference on Parallel and Distributed Computing Systems, October 1994.
- Member, technical program committee of 11th International Conference on Distributed Computing Systems, May 1991.
- Organizer, panel on "Distributed Shared Memory Systems" at 1996 IEEE International Conference on Computer Design: VLSI in Computers & Processors (ICCD '96), October 1996.
- Session Chair for: 19th International Conference on Parallel Processing, August 1990; 20th International Conference on Parallel Processing, August 1991; 22nd International Conference on Parallel Processing, August 1993; 5th IEEE Symposium on Parallel and Distributed Processing, December, 1993; 23rd International Conference on Parallel Processing, August 1994; 28th International Conference on Parallel Processing, September 1999; 20th IEEE Symposium on Reliable Distributed Systems (SRDS), October 2001.
- Invited Lecture to: Department of Electrical Engineering, Auburn University, May 1991; Department of Computer Science, University of Houston, March 1992; Department of Computer and Information Science, National Chiao-Tung University, Taiwan, May 1992; Department of Computer Science, National Tsing Hua University, Taiwan, May 1992; Department of Computer Science, Texas A&M University, November 1993; Department of Electrical and Computer Engineering, Iowa State University, March 1994; School of Electrical and Computer Engineering, Georgia Institute of Technology, May 1994; Department of Electrical and Computer Engineering, State University of New York at Buffalo, October 1995; University of Arkansas at Little Rock, September 1996; Department of Electrical Engineering, University of Texas at San Antonio, October 1996; Department of Computer Science, National Tsing Hwa University, Taiwan, April 2000.
- Referee, National Science Foundation (NSF).
- External Examiner, Nanyang Technological University, Singapore, 2001.
- Member, Evaluation Board of Istituto Superiore Mario Boella (ISMB, Italy).
- Referee, IEEE Transactions on Computers, IEEE Transactions on Parallel and Distributed Systems, IEEE/ACM Transactions on Networking, IEEE Transactions on Software Engineering, Journal of Parallel and Distributed Computing, IEEE Transactions on VLSI Systems, IEEE Transactions on Reliability, IEEE Transactions on Circuits and Systems, Part II, ACM Computing Reviews, Distributed Computing, Parallel Computing, Journal of Supercomputing, Information Processing Letters, Parallel Processing Letters, Journal of Information Science and Engineering.
- Referee, IEEE Computer magazine, IEEE Parallel & Distributed Technology: Systems and Applications, Computing Reviews.
- Referee, many international conferences and symposiums.

TEACHING EXPERIENCE

Course Developed

- CMPS 613 — Advanced Computer Communications and Networks
A new course I developed in 1999, with an emphasis on current technologies, standards and real implementations.
(Offered in Fall regularly since 1999)
- CMPS 513 - Computer Networks
This course was overhauled completely by N. Tzeng in 1997 when communication gear (including switches and wireless access points) and simulation tools (like OPNET and network simulators-2) were acquired and utilized for instructional use.
(Offered in Fall regularly since 1997)
- CMPS 530 — Principles of Computer Architecture
A substantial enhancement of this course was made by N. Tzeng for Spring 1996.
(Offered in Spring regularly since 1996)
- CMPS 631 — Advanced Parallel Processing
A major update of this course was performed by N. Tzeng for Spring 1995.
- CMPS 614 — Fault-Tolerant Computing
A new course developed by N. Tzeng in 1990, with coverage on basic concepts, design methodologies, and implementations of fault-tolerant computing systems.
(Offered in Fall 1990 - 1994, Fall 1996)
- CMPS 639 — Advanced Topics in VLSI/WSI-Based Reliable Computing Systems
A new course designed and offered together with L. Bhuyan in Fall 1988, covering foremost research on yield and reliability modeling, reliable VLSI/WSI-based system layouts, and system reconfiguration.

Courses Offered

- CMPS 430 — Computer Architecture
(Spring 1994 - Spring 1995)
- CMPS 530 — Principles of Computer Architecture
(Spring 1996 - Present)
- CMPS 513 — Computer Networks
(Fall 1997 - Fall 1998)
- CMPS 613 — Advanced Computer Communications and Networks
(Fall 1999, Fall 2001 - Present)
- CMPS 533 — Distributed Computing Systems
(Fall 1988)
- CMPS 531 — Parallel Processing
(Spring 1989 - Spring 1990)
- CMPS 631 — Advanced Parallel Processing
(Spring 1995)
- CMPS 614 — Fault-Tolerant Computing
(Fall 1990 - 1994, Fall 1996)

- CMPS 583 — Computer Design and Implementation
(Spring 1989 - Spring 1994)
- CMPS 639 — Advanced Topics on VLSI/WSI Computing Systems
(Fall 1988)

Research Laboratory Establishment

With funding from National Science Foundation (through a competitive MRI grant) and Board of Regents, State of Louisiana, I have established Computer Architecture and Networks (CAN) Laboratory at Center for Advanced Computer Studies (CACS). CAN houses an array of the state-of-the-art computing and communication equipment dedicated to the support of research projects and educational activities. It includes some 20 SUN (Ultra-5 and Ultra-10) workstations and a SUN Enterprise E-420 (4-way) server, interconnected by fast Ethernet switches and Gigabit Ethernet switches. About two dozens of desktop PC's and laptops, plus one Alpha 21264 workstation, are also interconnected through the fast Ethernet switches and three wireless access points. Various peripheral devices and (application and system) software are available for research use. These machines constitute a heterogeneous environment and offer different platforms indispensable for our research on various topics in computer networks and computer architecture.

Ph.D. Comprehensive Examination Overhaul

I was in charge of a complete overhaul of the Ph.D. comprehensive exam system in Spring 1996, streamlining subjects into fewer areas of concentration and requiring the exam to comprise two parts: breadth requirement (by course work) and written examination requirement (of 2 chosen areas). This comprehensive system has been adopted in CACS since Fall 1996, encouraging students to focus more on research work and successfully leading to more student publications.

Ph.D. Dissertations Supervised

- Afshin Ganjoo, Ph.D., Computer Science, in May 2000, dissertation title: New Techniques for Minimizing Branch Penalties in Pipelined Processors.
(Current Position: Research Staff, HP Labs, Hewlett-Packard Company, California)
- Angkul Kongmunvattana, Ph.D., Computer Science, in Dec. 1999, dissertation title: Coherence-Based Fault Tolerance for Software Distributed Shared Memory.
(Current Position: Assistant Professor, Department of Computer Science, University of Nevada, Reno)
- Shiwa S. Fu, Ph.D., Computer Science, in May 1997, dissertation title: Aggressive Release Consistency and Efficient Lock Implementation for Software Distributed Shared Memory.
(Current Position: Member of Technical Staff, IBM Thomas J. Watson Research Center, Yorktown Heights, New York)
- Guanghua Ling, Ph.D., Computer Engineering, in May 1995, dissertation title: Reconfiguration and Performance of Distributed-Memory Parallel Systems.
(Current Position: Staff Member, IBM Global Services Center)
- Chung-Han Chen, Ph.D., Computer Engineering, in March 1993, dissertation title: Yield and Reliability Aspects of VLSI/WSI Parallel Processing Architectures.
(Current Position: Assistant Professor, Department of Computer Science, Tuskegee University, Alabama)
- Po-Jen Chuang, Ph.D., Computer Science, in June 1992, dissertation title: Processor Resource Management in Partitionable Parallel Computers.

(Current Position: Professor and Chair, Department of Electrical Engineering, TanKang University, Taiwan)

- Kemathat Vibhatavanij, Computer Science, expected in May 2002, dissertation title: Design and Evaluation of High-Performance Routers.

M.S. Graduates Supervised

- Malcolm Mandviwalla, M.S., Computer Science, in August 2001, master thesis title: A Cost-Effective Switching Fabric for Scalable Routers.
- Prasanth Alla, M.S., Computer Science, expected in December 2001, project title: Implementation of Scalable Multicast Based on Shared Structures Using the *ns-2* Simulator.
- Santipong Tanchatchawal, M.S., Computer Science, in May 2000, master thesis title: Checkpointing Techniques for Home-Less and Home-Based Software Distributed Shared Memory Systems.
- Laxminarayana Tumuluru, M.S., Computer Science, in Dec. 1999, project title: Performance Assessment of Multicast Routing Algorithms in Ad Hoc Networks Using Network Simulator *ns-2*.
- Steven Noel, M.S., Computer Engineering, in Dec. 1998, project title: Wavelets and Neural Networks for Radar Target Detection and Velocity Estimation.
- Afshin Ganjoo, M.S., Computer Science, in Dec. 1997, thesis title: Branch Assertion and Branch Elimination for High-Performance Processors.
- Kiran Ponnuru, M.S., Computer Science, in Dec. 1997, thesis title: Cost-Effective Switching Architecture for High-Speed Networks.
- Mangaprasad R. Gottam, M.S., Computer Engineering, in Dec. 1997, project title: VLSI Implementation of Highly Parallel System Architecture.
- Angkul Kongmunvattana, M.S., Computer Science, in May 1997, master thesis title: Performance Issues in Software Distributed Shared Memory Systems.
- Balakrishna D. Jayadev, M.S., Computer Engineering, in May 1997, project title: Design and Evaluation of a High-Performance Parallel Interface (HiPPI).
- Lei Yu, M.S., Computer Science, in May 1995, project title: Analytic Study of a RAID (Redundant Array of Inexpensive Disks) Design.
- Buchamwandla Ravinandan, M.S., Computer Engineering, in May 1994, project title: Efficient Adaptive Routing in 3-D Meshes.
- Jun Yan, M.S., Computer Science, in May 1994, project title: A Declustering Scheme for Parity Stripes in Redundant Disk Arrays.
- Supamas Sirichotiyakul, M.S., Computer Science, in Dec. 1993, project title: Improving Execution Performance of Parallel Codes on the Intel *iPSC/860*.
- Junlin Tang, M.S., Computer Science, in Dec. 1993, project title: A Reliable Interconnection Network Composed of 3×3 Switches.
- Venkata R. Kalapatapu, M.S., Computer Science, in Dec. 1993, project title: Analytic model of Multi-Queues Switches.
- Yu-Yun Liao, M.S., Computer Engineering, in May 1993, project title: The Palindrome Network for Fault-Tolerant Interconnection.
- Bhagavathi R. Mula, M.S., Computer Engineering, in May 1992, project title: Efficient Task Migration in Hypercubes.
- Wendy M. Auzenne, M.S., Computer Engineering, in May 1992, project title: Performance Analysis of Multistage Interconnection Networks with Multi-Queues under Nonuniform Traffic.
- Kalyanakri Ramanathan, M.S., Computer Engineering, in May 1991, project title: Performance Improvement of Multistage Interconnection Networks Using Feedback Control.

- Venkat R. Podduturi, M.S., Computer Engineering, in Dec. 1990, project title: A Yield Enhancement Reconfiguration Algorithm for VLSI Processor Arrays with Clustering Defects.
- Pardha V. Pothana, M.S., Computer Engineering, in Dec. 1990, project title: I/O Embedding in Hypercubes.
- Pravin Nadkarni, M.S., Computer Engineering, in Dec. 1989, project title: A Low Cost Multistage Network Switch for Relaxing Tree Saturation Effects.
- Sizheng Wei, M.S., Computer Science, in Aug. 1988, project title: Hypercubes with Improved Performance.

Ph.D. Dissertation Committees Served

Mohammad Bassem Maaz, Computer Engineering, 1999; Sanjeev Ramakumar, Computer Science, 1999; Mohammad-Mazen Al-Khatib, Computer Engineering, 1998; Amr A. Elchouemi, Computer Engineering, 1997; Jimmy Chan Limqueco, Computer Engineering, 1997; Bushord B. Thomas, Computer Science, 1997; Georgios A. Demetriou, Computer Engineering, 1997; Ramesh Kolluru, Computer Science, 1996; Paul Edward Shipley, Computer Engineering, 1996; George Paschos, Computer Science, 1996; Rafic Ayoubi, Computer Engineering, 1994; Kanonkluk Vanapipat, Computer Science, 1994; Raghava V. Cherabuddi, Computer Engineering, 1994; Bassem A. AlHalabi, Computer Engineering, 1994; Antonio Fernandez, Computer Science, 1994; N. A. Ramakrishna, Computer Science, 1993; Aakash Tyagi, Computer Engineering, 1992; Ay-Hwa Andy Liou, Computer Science, 1991.

UNIVERSITY COMMITTEE SERVICE

- Chair, Graduate Faculty Peer Review Committee (1999 - 2000)
- Member, College of Sciences, Dean Search Committee (2001 - 2002)
- Member, Honors Convocation Committee (2001 - Present)
- Member, Graduate Faculty Peer Review Committee (1997 - 2000)

DEPARTMENTAL COMMITTEE SERVICE

- Chair, Faculty Recruitment Committee (2000 - 2001)
- Chair, Ph.D. Comprehensive Exam (1996 - 2000)
- Chair, Ph.D. Comprehensive Exam Evaluation and Recommendation Committee (1995 - 1996)
- Chair, Ph.D. Comprehensive Exam Committee on Computer Architecture (1995 - Present)
- Chair, Ph.D. Comprehensive Exam Committee on Computer Design and Implementation (1989 - 1994)
- Co-Chair, Faculty Recruitment Committee (1990 - 1991)
- Member, Faculty Recruitment Committee (2000 - Present)
- Member, Ph.D. Comprehensive Exam Committee on Computer Design & Implementation, and VLSI (1996 - Present)
- Member, Student Admissions and Financial Support Committee (1994 - 1999)
- Member, Ph.D. Comprehensive Exam Committee on Computer Design and Implementation (1989 - 1996)
- Member, Student Affairs Committee (1995 - 1997)
- Member, Graduate Studies Committee (1987 - 1990)
- Member, Research Committee (1990 - 1994)